MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, AND, DIGITAL, KEY CODE DQJ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic AND gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

* 3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 20260-20T16:55Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.052 failures per million operating hours for class I or 0.081 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DQJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/244-1 for class I and M28787/244-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

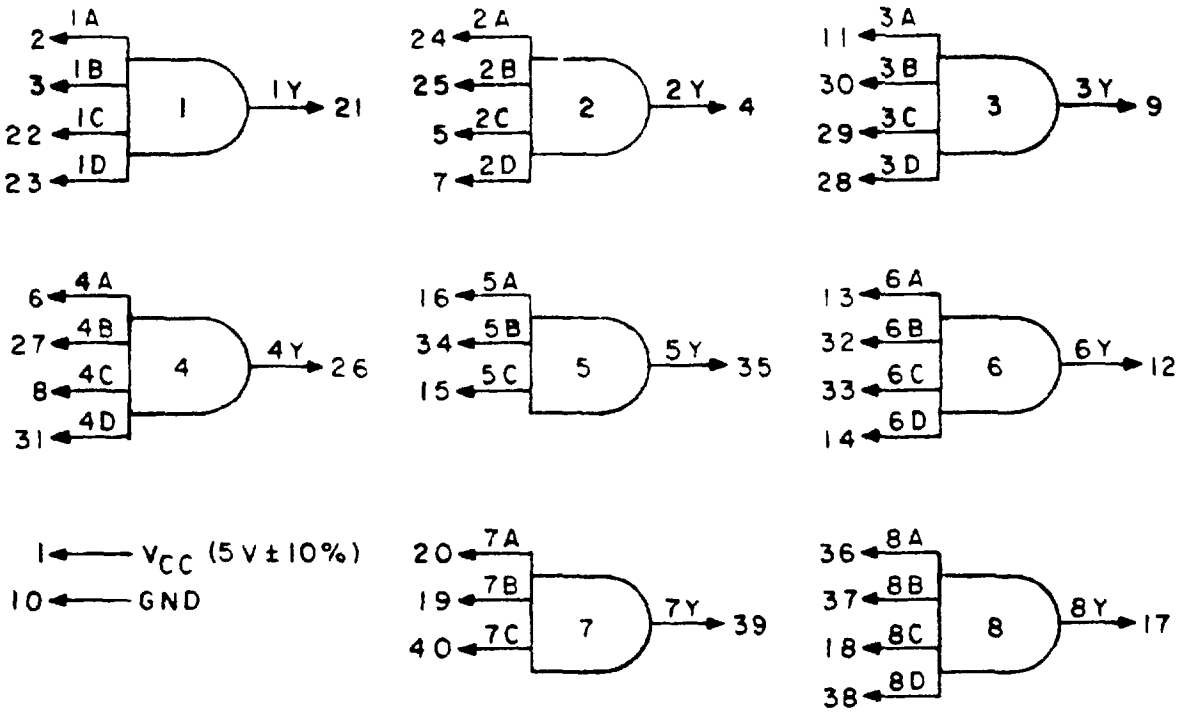


FIGURE 1. DQJ module function diagram, AND gate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Frame Isolation													
	Initial	III	S	10	[illegible]	10	[illegible]	10	[illegible]	10	[illegible]	10	[illegible]	MΩ
	End-of-life	III	S	1	[illegible]	1	[illegible]	1	[illegible]	1	[illegible]	1	[illegible]	MΩ
B1-B8	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C30	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D30	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E8	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F30	Output voltage, low level	VII	X	0	400	0	400	0	400	0	400	0	400	mV
G1-G30	Input current, high level	VIII	X	0	50	0	50	0	50	0	50	0	50	µA
H1-H30	Input current, low level	IX	X	-0.10	-2.18	-0.10	-2.18	-0.10	-2.18	-0.10	-2.18	-0.10	-2.18	mA
J1-J8	Output short circuit current	X	X	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
K1	Power supply current, low level	XI	S	1	128	1	128	1	128	1	128	1	128	mA
K2	Power supply current, high level	XI	S	1	80	1	80	1	80	1	80	1	80	mA
L1-L8	Transition time (t_{TLH}), C_L = 50 pF	XII	S	3	15	3	15	3	20	3	15	3	25	ns
L9-L16	Transition time (t_{TLH}), C_L = 150 pF	XII	Q	3	25	3	25	3	30	3	25	3	35	ns
M1-M8	Transition time (t_{THL}), C_L = 50 pF	XIII	S	3	15	3	15	3	20	3	15	3	25	ns
M9-M16	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	3	25	3	25	3	30	3	25	3	35	ns
N1-N30	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	3	15	3	15	3	20	3	15	3	25	ns
N31-N60	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	3	25	3	25	3	30	3	25	3	35	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:55Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
[illegible]	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	3	15	3	15	3	20	3	15	3	25	ns
[illegible]	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	3	25	3	25	3	30	3	25	3	35	ns
[illegible]	Output crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
[illegible]	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
[illegible]	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater), at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:55Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through U shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

Supply voltage	5 V dc ±10%		
	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	704	704	mW
Input current (for four-input gates)			
Low level (V_{IL} = 0 V)	-2.18 max	-2.18 max	mA
(V_{IL} = 0.4 V)	-2 max	-2 max	mA
High level (V_{IH} = 2 V)	50 max	50 max	µA
(V_{IH} = 5.5 V)	1 max	1 max	mA

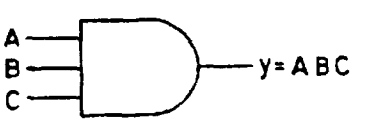
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

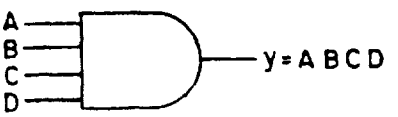
	Class I	Class II	Unit
Input current (for three-input gates)			
Low level (V_{IL} = 0 V)	-2.18 max	-2.18 max	mA
(V_{IL} = 0.4 V)	-2 max	-2 max	mA
High level (V_{IH} = 2 V)	50 max	50 max	µA
(V_{IH} = 5.5 V)	1 max	1 max	mA
Input threshold voltage			
Low level	-0.5 min, 0.8 max	-0.5 min, 0.8 max	V
High level	2 min, V_{CC} max	2 min, V_{CC} max	V
Output voltage			
Low level (I_{OL} = 20 mA)	0.4 max	0.4 max	V
High level (I_{OH} = -0.5 mA)	2.4 min	2.4 min	V
Output drive current			
Low level (V_{OL} = 0.4 V)	20	20	mA
High level (V_{OH} = 2.4 V)	-0.5	-0.5	mA

* 6.5 Module function. This module contains six 4-input and two 3-input high speed AND gates with totem-pole outputs.

6.5.1 Input to output relationship. Each AND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.





Inputs	Outputs
Any input in low state	Low state
All inputs in high state	High state

6.5.2 Parallel outputs. Paralleling of outputs is not permissible, except when inputs and outputs of the same IC are paralleled to obtain increased drive. Each circuit whose output is paralleled with an output of another circuit must have each of its inputs paralleled with a corresponding input of the other circuit. The circuit groups that may be paralleled are as follows:

a. Circuits 1 and 2.

b. Circuits 3 and 4.

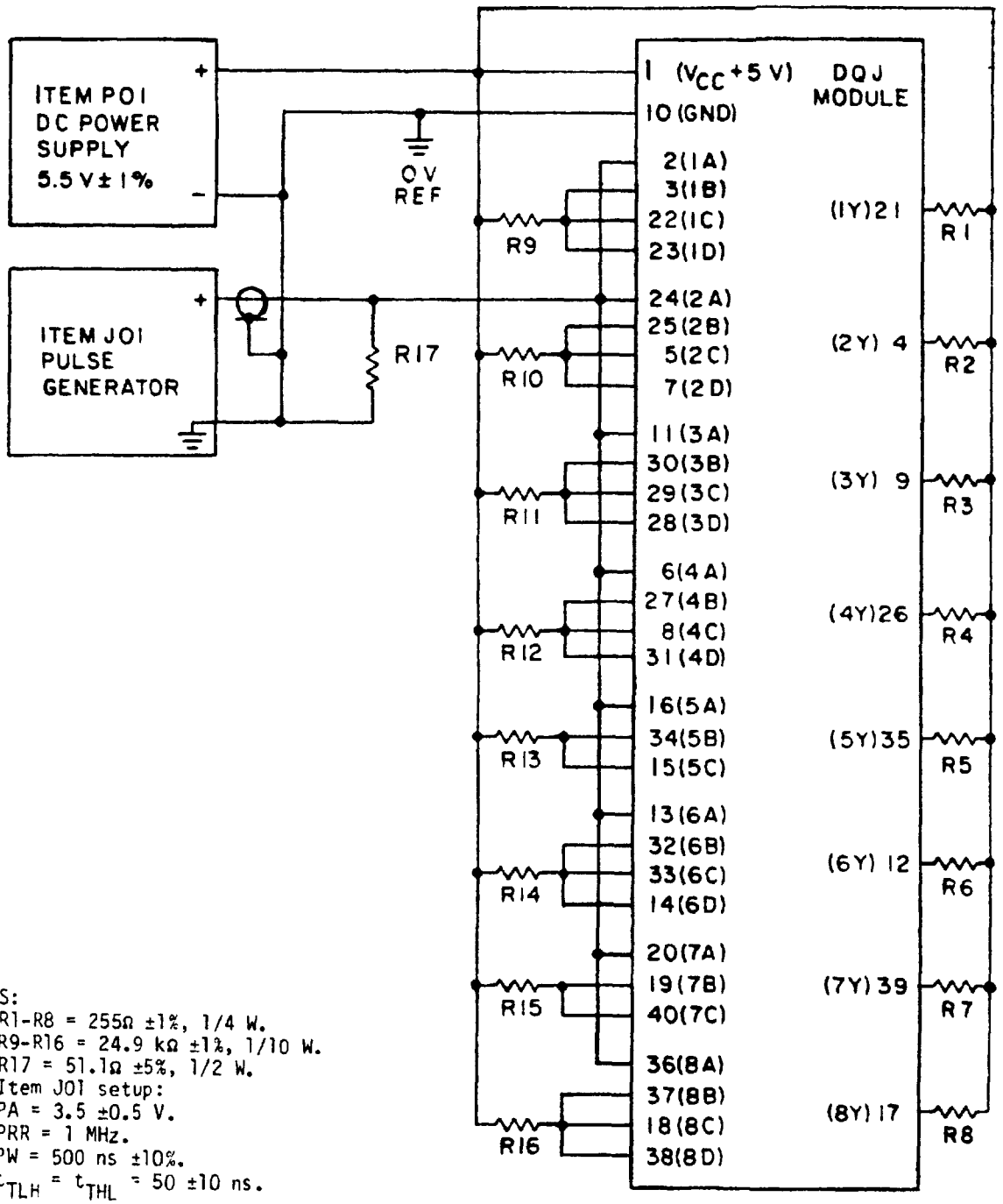
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N517)

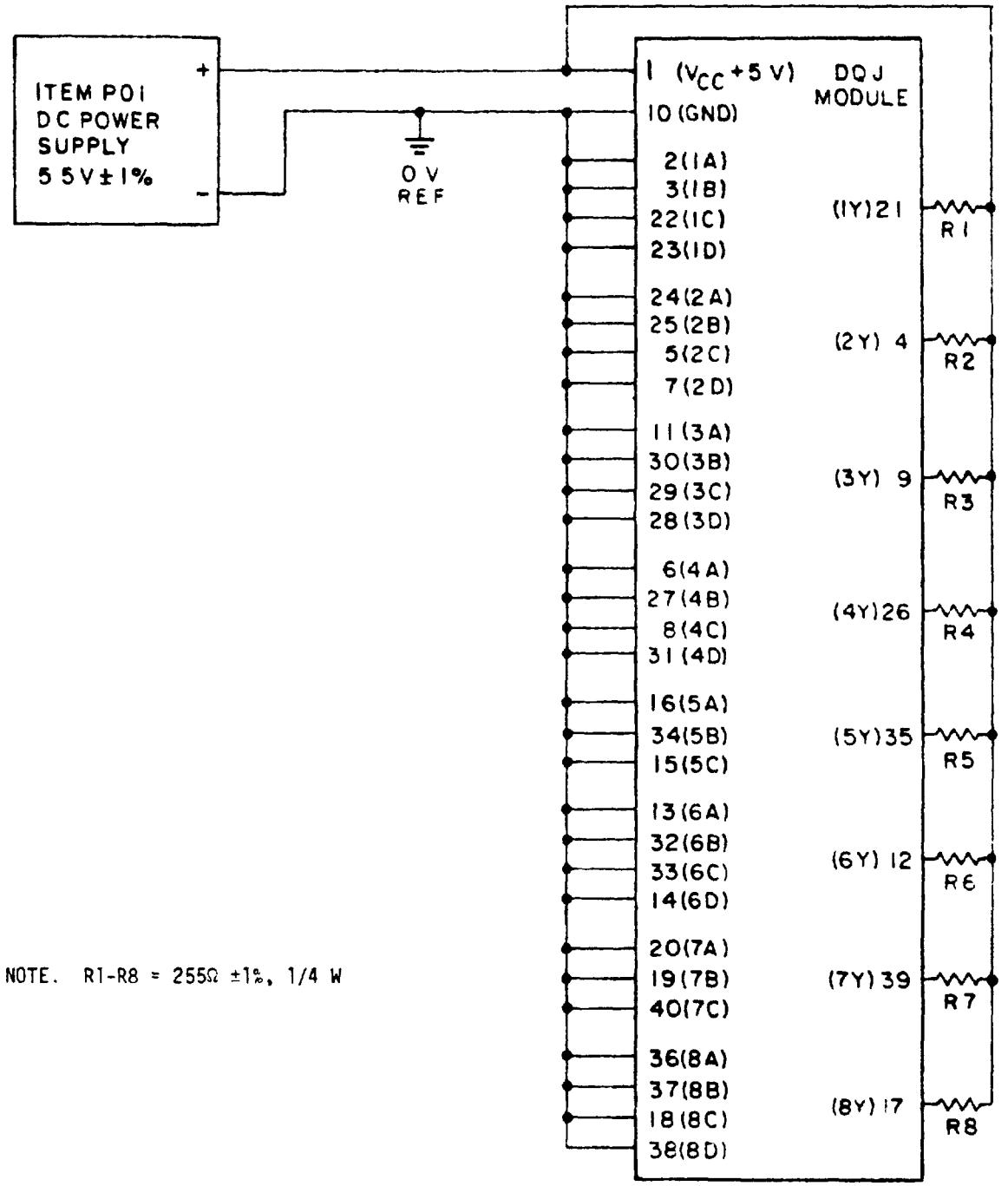
Source: https://assist.dla.mil -- Downloaded: 202640-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R8 = 255Ω ±1%, 1/4 W.
2. R9-R16 = 24.9 kΩ ±1%, 1/10 W.
3. R17 = 51.1Ω ±5%, 1/2 W.
4. Item J01 setup:
   PA = 3.5 ±0.5 V.
   PRR = 1 MHz.
   PW = 500 ns ±10%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

* FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTE. R1-R8 = 255Ω ±1%, 1/4 W

FIGURE 3 Component temperature test circuit

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.

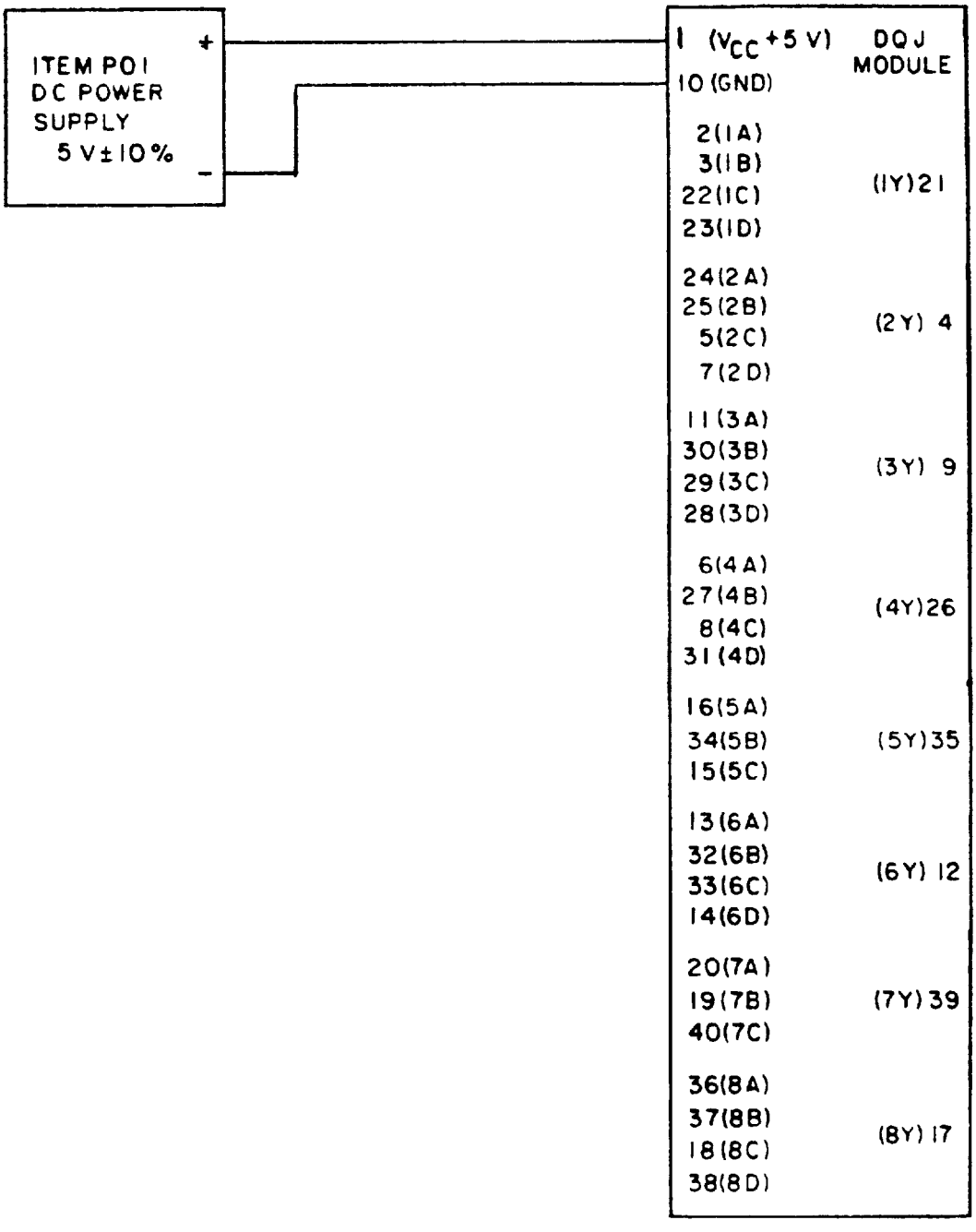


FIGURE 4 Vibration test circuit

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.

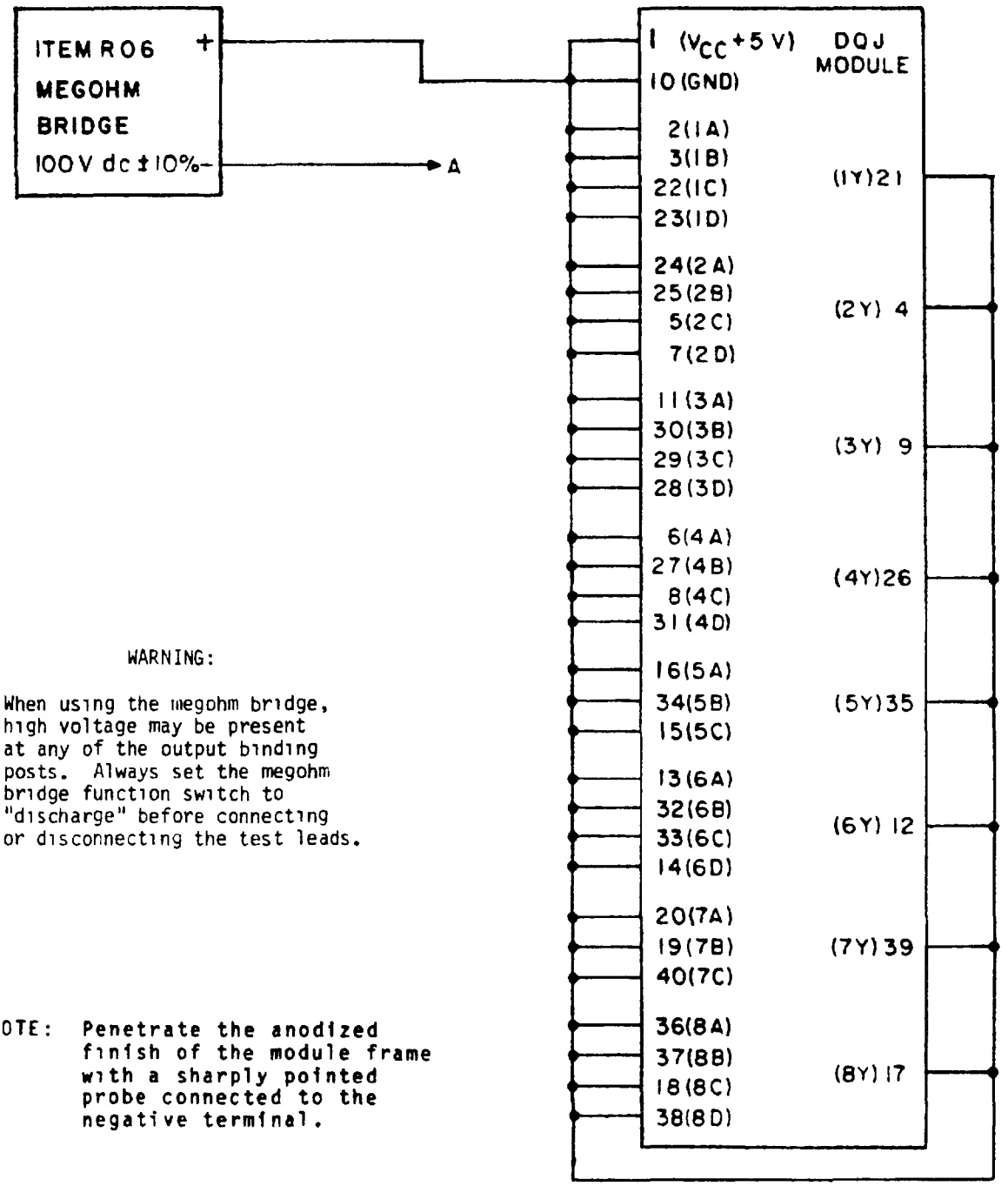
TABLE III. Frame isolation test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A	Frame isolation	Module frame	10	∞	MΩ	5

*TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	V_{OUT} (low loaded)	21	10	800	mV	6
B2	"	4	"	"	"	"
B3	"	9	"	"	"	"
B4	"	26	"	"	"	"
B5	"	35	"	"	"	"
B6	"	12	"	"	"	"
B7	"	39	"	"	"	"
B8	"	17	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T16:55Z
Check the source to verify that this is the current version before use.



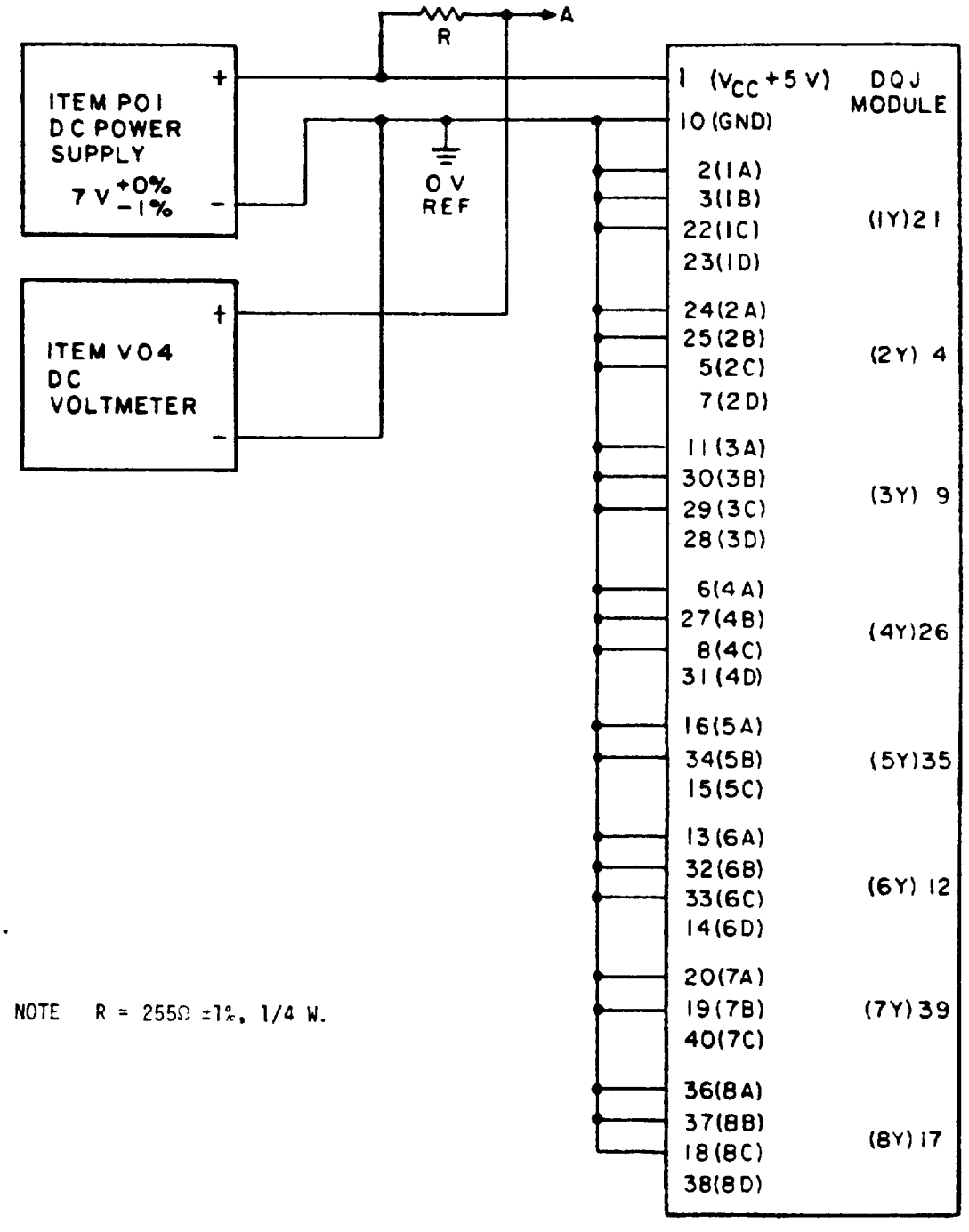
WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 5. Frame isolation test for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.



NOTE R = 255Ω ±1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

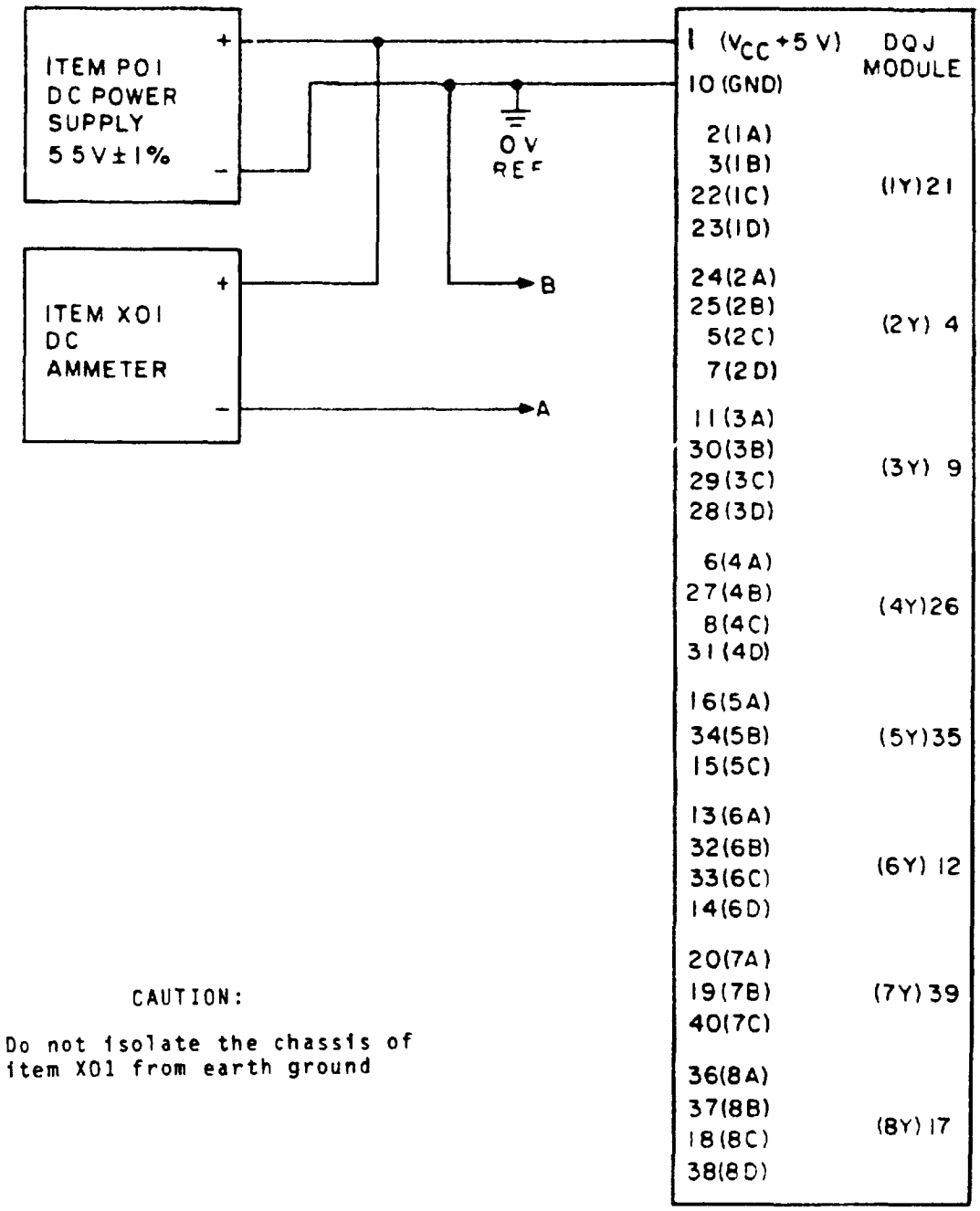
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	2	3,22,23	0	1	mA	7
C2	1B "	3	2,22,23	"	"	"	"
C3	1C "	22	2,3,23	"	"	"	"
C4	1D "	23	2,3,22	"	"	"	"
C5	2A "	24	25,5,7	"	"	"	"
C6	2B "	25	24,5,7	"	"	"	"
C7	2C "	5	24,25,7	"	"	"	"
C8	2D "	7	24,25,5	"	"	"	"
C9	3A "	11	30,29,28	"	"	"	"
C10	3B "	30	11,29,28	"	"	"	"
C11	3C "	29	11,30,28	"	"	"	"
C12	3D "	28	11,30,29	"	"	"	"
C13	4A "	6	27,8,31	"	"	"	"
C14	4B "	27	6,8,31	"	"	"	"
C15	4C "	8	6,27,31	"	"	"	"
C16	4D "	31	6,27,8	"	"	"	"
C17	5A "	16	34,15	"	"	"	"
C18	5B "	34	16,15	"	"	"	"
C19	5C "	15	16,34	"	"	"	"
C20	6A "	13	32,33,14	"	"	"	"
C21	6B "	32	13,33,14	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C22	6C Input breakdown current	33	13,32,14	0	1	mA	7
C23	6D "	14	13,32,33	"	"	"	"
C24	7A "	20	19,40	"	"	"	"
C25	7B "	19	20,40	"	"	"	"
C26	7C "	40	20,19	"	"	"	"
C27	8A "	36	37,18,38	"	"	"	"
C28	8B "	37	36,18,38	"	"	"	"
C29	8C "	18	36,37,38	"	"	"	"
C30	8D "	38	36,37,18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground

FIGURE 7 Input breakdown current test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test.

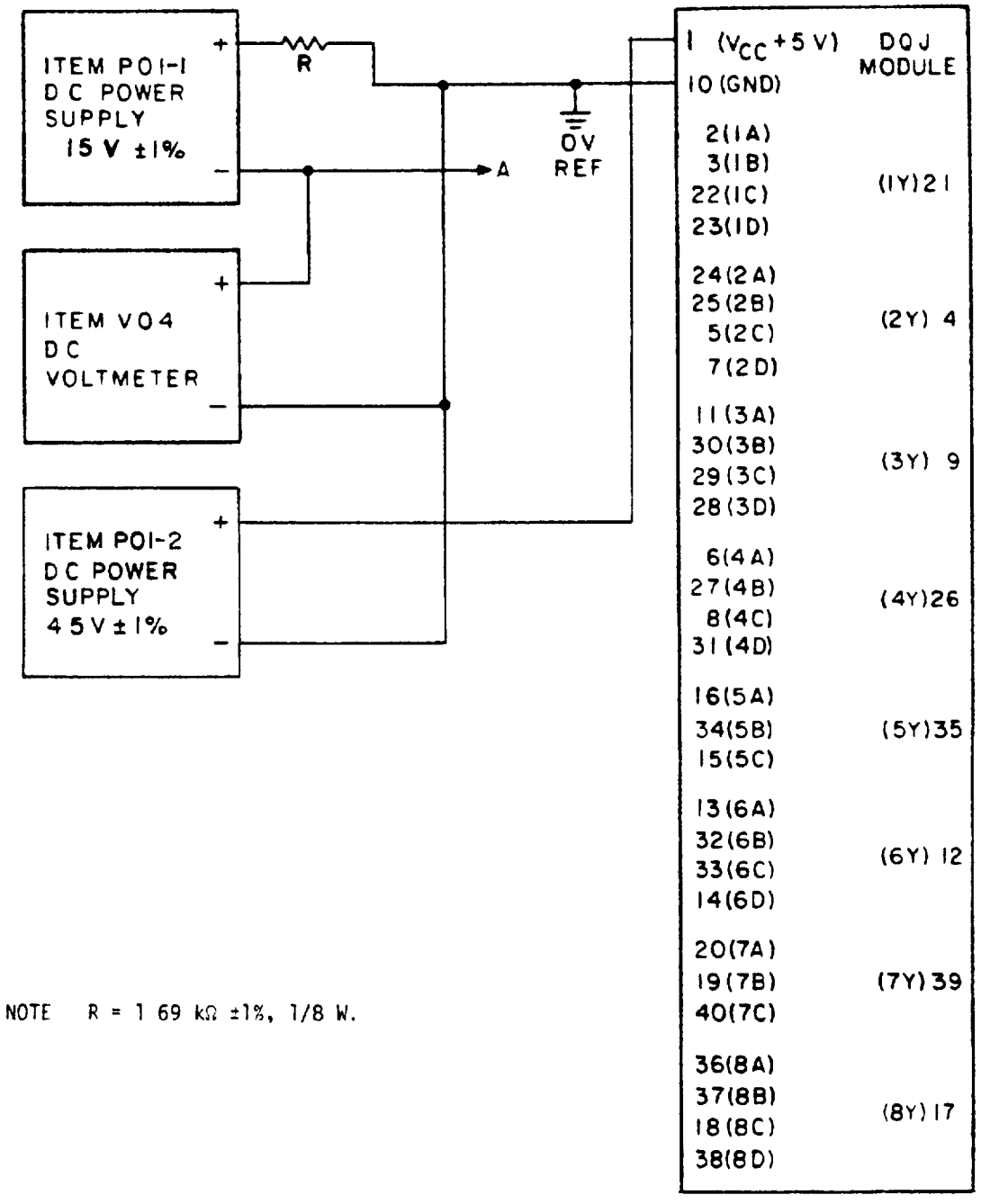
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1A Input clamp diode voltage	2	-0.4	-1.5	V	8
D2	1B "	3	"	"	"	"
D3	1C "	22	"	"	"	"
D4	1D "	23	"	"	"	"
D5	2A "	24	"	"	"	"
D6	2B "	25	"	"	"	"
D7	2C "	5	"	"	"	"
D8	2D "	7	"	"	"	"
D9	3A "	11	"	"	"	"
D10	3B "	30	"	"	"	"
D11	3C "	29	"	"	"	"
D12	3D "	28	"	"	"	"
D13	4A "	6	"	"	"	"
D14	4B "	27	"	"	"	"
D15	4C "	8	"	"	"	"
D16	4D "	31	"	"	"	"
D17	5A "	16	"	"	"	"
D18	5B "	34	"	"	"	"
D19	5C "	15	"	"	"	"
D20	6A "	13	"	"	"	"
D21	6B "	32	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D22	6C Input clamp diode voltage	33	-0.4	-1.5	V	8
D23	6D "	14	"	"	"	"
D24	7A "	20	"	"	"	"
D25	7B "	19	"	"	"	"
D26	7C "	40	"	"	"	"
D27	8A "	36	"	"	"	"
D28	8B "	37	"	"	"	"
D29	8C "	18	"	"	"	"
D30	8D "	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2024-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTE R = 1 69 kΩ ±1%, 1/8 W.

FIGURE 8 Input clamp diode voltage test circuit for table VI

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.

*TABLE VII Output voltage (high-level and low-level) test.

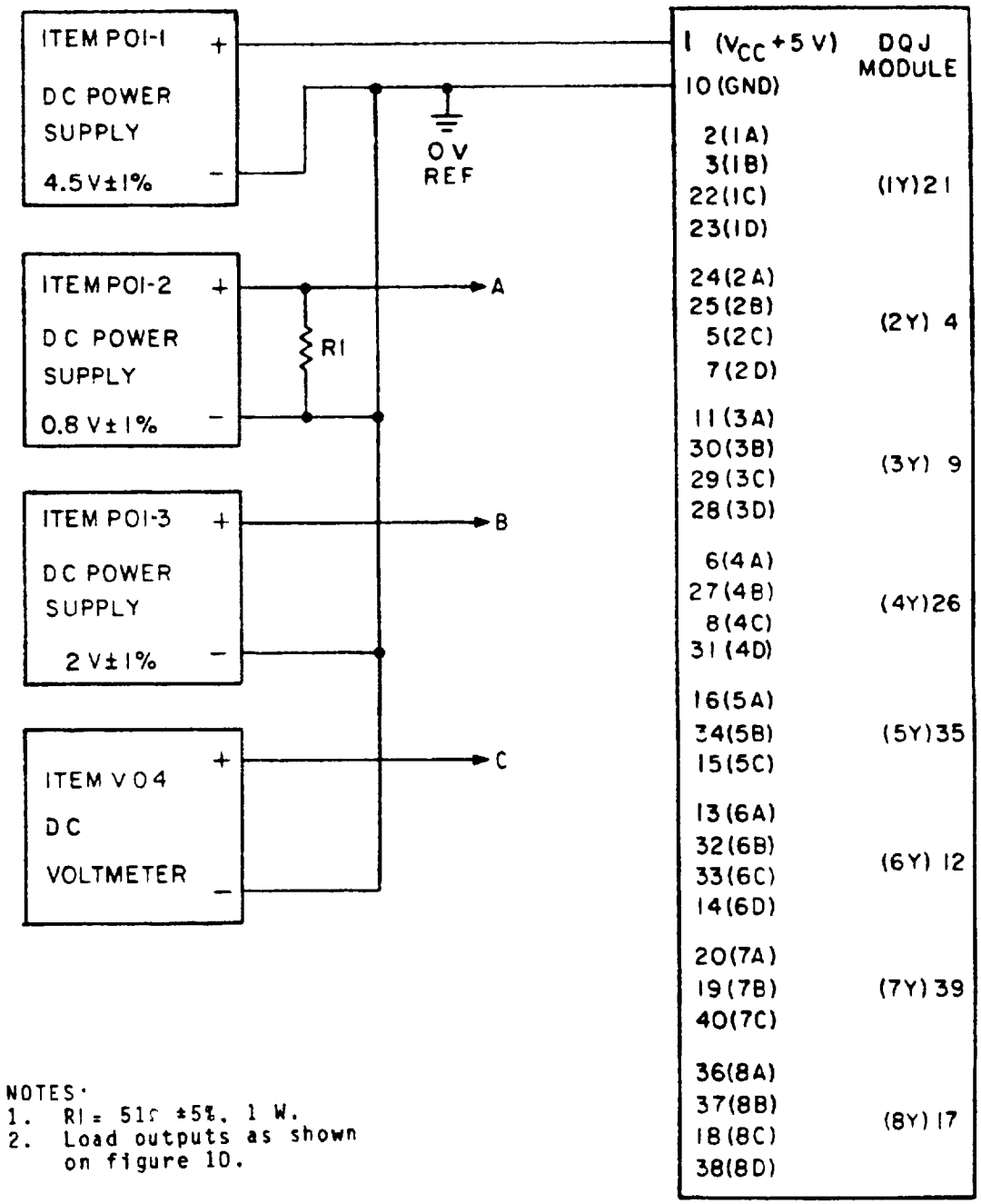
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1Y Output voltage, high level	Open	2,3,22,23	21	2.4	4.5	V	9
E2	2Y "	"	24,25,5,7	4	"	"	"	"
E3	3Y "	"	11,30,29,28	9	"	"	"	"
E4	4Y "	"	6,27,8,31	26	"	"	"	"
E5	5Y "	"	16,34,15	35	"	"	"	"
E6	6Y "	"	13,32,33,14	12	"	"	"	"
E7	7Y "	"	20,19,40	39	"	"	"	"
E8	8Y "	"	36,37,18,38	17	"	"	"	"
F1	1Y Output voltage, low level	3,22,23	2	21	0	400	mV	"
F2	1Y "	2,22,23	3	"	"	"	"	"
F3	1Y "	2,3,23	22	"	"	"	"	"
F4	1Y "	2,3,22	23	"	"	"	"	"
F5	2Y "	25,5,7	24	4	"	"	"	"
F6	2Y "	24,5,7	25	"	"	"	"	"
F7	2Y "	24,25,7	5	"	"	"	"	"
F8	2Y "	24,25,5	7	"	"	"	"	"
F9	3Y "	30,29,28	11	9	"	"	"	"
F10	3Y "	11,29,28	30	"	"	"	"	"
F11	3Y "	11,30,28	29	"	"	"	"	"
F12	3Y "	11,30,29	28	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

*TABLE VII. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F13	4Y Output voltage, low level	27,8,31	6	26	0	400	mV	9
F14	4Y "	6,8,31	27	"	"	"	"	"
F15	4Y "	6,27,31	8	"	"	"	"	"
F16	4Y "	6,27,8	31	"	"	"	"	"
F17	5Y "	34,15	16	35	"	"	"	"
F18	5Y "	16,15	34	"	"	"	"	"
F19	5Y "	16,34	15	"	"	"	"	"
F20	6Y "	32,33,14	13	12	"	"	"	"
F21	6Y "	13,33,14	32	"	"	"	"	"
F22	6Y "	13,32,14	33	"	"	"	"	"
F23	6Y "	13,32,33	14	"	"	"	"	"
F24	7Y "	19,40	20	39	"	"	"	"
F25	7Y "	20,40	19	"	"	"	"	"
F26	7Y "	20,19	40	"	"	"	"	"
F27	8Y "	37,18,38	36	17	"	"	"	"
F28	8Y "	36,18,38	37	"	"	"	"	"
F29	8Y "	36,37,38	18	"	"	"	"	"
F30	8Y "	36,37,18	38	"	"	"	"	"

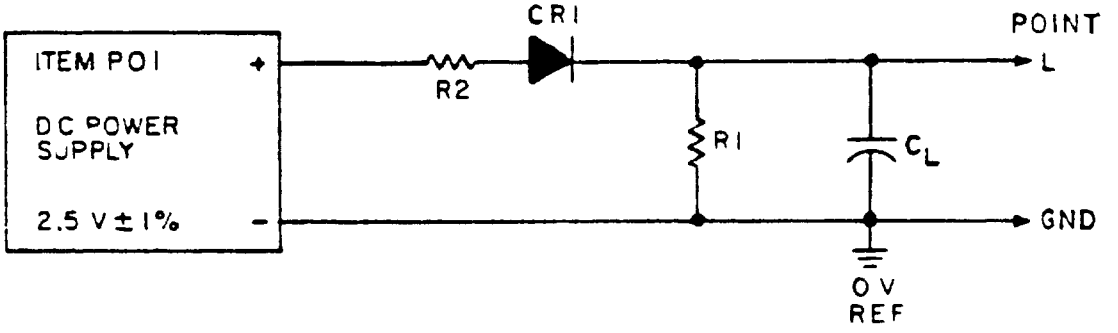
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load outputs as shown on figure 10.

* FIGURE 9. Output voltage (high-level and low-level) test circuit for table VII.

Source: https://assist.dla.mil Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES
1. CR1 = 1N4150 or equivalent.
2. R1 = 4.87 kΩ ±1%, 1/8 W.
3. R2 = Select value for current flow of 20 mA ±1% out of point L with point L held at 0.4 V.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions L1-L8, M1-M8, N1-N30, P1-P30, Q1-Q8, and R1-R8.
5. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions L9-L16, M9-M16, N31-N60, and P31-P60.
6. C_L is not required for test conditions E and F.

FIGURE 10. DQJ logic load circuit

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (high-level) test

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1A Input current, high level	2	Open	3,22,23	0	50	µA	11
G2	1B "	3	"	2,22,23	"	"	"	"
G3	1C "	22	"	2,3,23	"	"	"	"
G4	1D "	23	"	2,3,22	"	"	"	"
G5	2A "	24	"	25,5,7	"	"	"	"
G6	2B "	25	"	24,5,7	"	"	"	"
G7	2C "	5	"	24,25,7	"	"	"	"
G8	2D "	7	"	24,25,5	"	"	"	"
G9	3A "	11	"	30,29,28	"	"	"	"
G10	3B "	30	"	11,29,28	"	"	"	"
G11	3C "	29	"	11,30,28	"	"	"	"
G12	3D "	28	"	11,30,29	"	"	"	"
G13	4A "	6	"	27,8,31	"	"	"	"
G14	4B "	27	"	6,8,31	"	"	"	"
G15	4C "	8	"	6,27,31	"	"	"	"
G16	4D "	31	"	6,27,8	"	"	"	"
G17	5A "	16	"	34,15	"	"	"	"
G18	5B "	34	"	16,15	"	"	"	"
G19	5C "	15	"	16,34	"	"	"	"
G20	6A "	13	"	32,33,14	"	"	"	"
G21	6B "	32	"	13,33,14	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G22	6C Input current, high level	33	Open	13,32,14	0	50	µA	11
G23	6D "	14	"	13,32,33	"	"	"	"
G24	7A "	20	"	19,40	"	"	"	"
G25	7B "	19	"	20,40	"	"	"	"
G26	7C "	40	"	20,19	"	"	"	"
G27	8A "	36	"	37,18,38	"	"	"	"
G28	8B "	37	"	36,18,38	"	"	"	"
G29	8C "	18	"	36,37,38	"	"	"	"
G30	8D "	38	"	36,37,18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE IX Input current (low-level) test

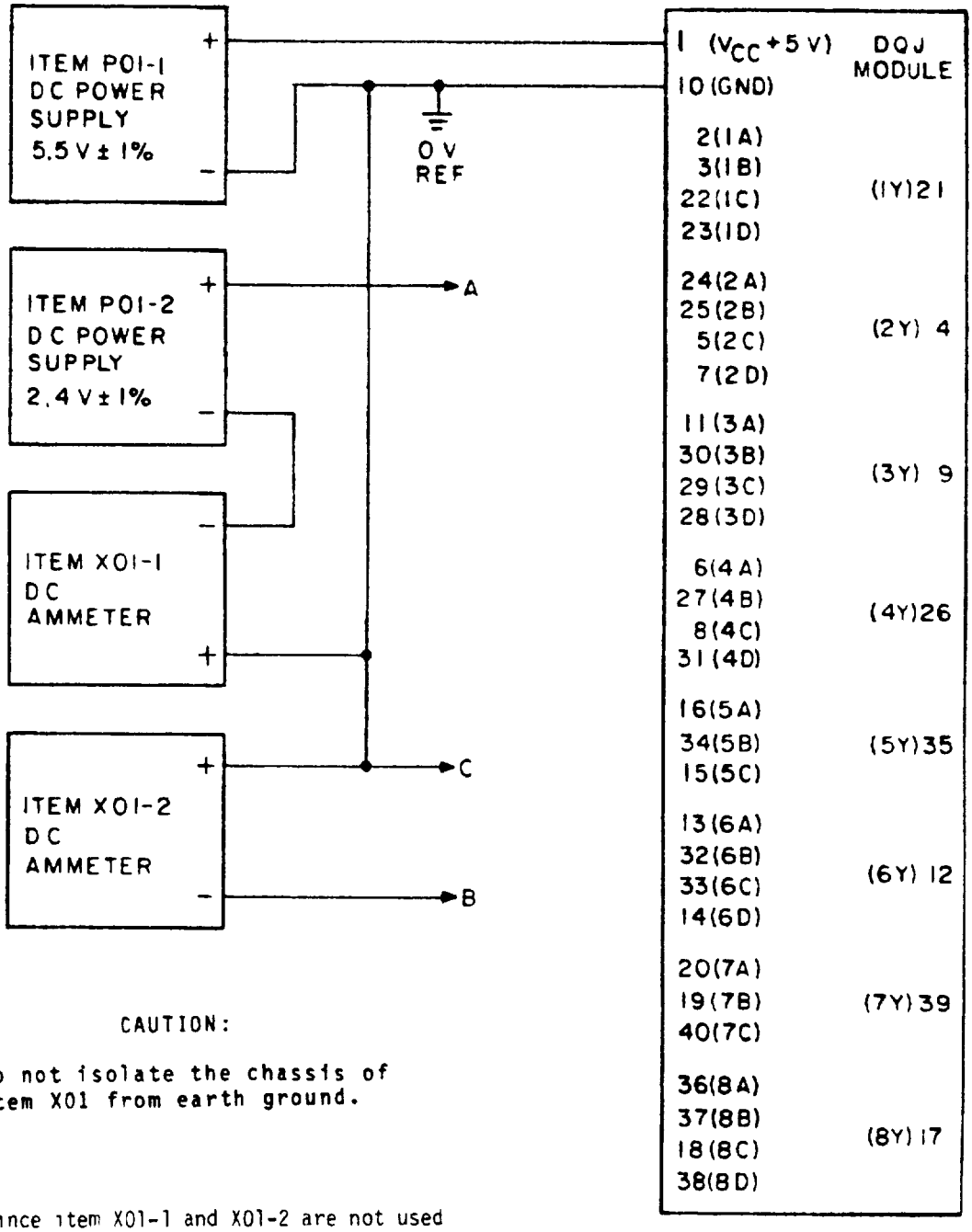
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1A Input current, low level	Open	2	3,22,23	-0.10	-2.18	mA	11
H2	1B "	"	3	2,22,23	"	"	"	"
H3	1C "	"	22	2,3,23	"	"	"	"
H4	1D "	"	23	2,3,22	"	"	"	"
H5	2A "	"	24	25,5,7	"	"	"	"
H6	2B "	"	25	24,5,7	"	"	"	"
H7	2C "	"	5	24,25,7	"	"	"	"
H8	2D "	"	7	24,25,5	"	"	"	"
H9	3A "	"	11	30,29,28	"	"	"	"
H10	3B "	"	30	11,29,28	"	"	"	"
H11	3C "	"	29	11,30,28	"	"	"	"
H12	3D "	"	28	11,30,29	"	"	"	"
H13	4A "	"	6	27,8,31	"	"	"	"
H14	4B "	"	27	6,8,31	"	"	"	"
H15	4C "	"	8	6,27,31	"	"	"	"
H16	4D "	"	31	6,27,8	"	"	"	"
H17	5A "	"	16	34,15	"	"	"	"
H18	5B "	"	34	16,15	"	"	"	"
H19	5C "	"	15	16,34	"	"	"	"
H20	6A "	"	13	32,33,14	"	"	"	"
H21	6B "	"	32	13,33,14	"	"	"	"

Source: https://assist.dla.mil 26 Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H22	6C Input current, low level	Open	33	13,32,14	-0.10	-2.18	mA	11
H23	6D "	"	14	13,32,33	"	"	"	"
H24	7A "	"	20	19,40	"	"	"	"
H25	7B "	"	19	20,40	"	"	"	"
H26	7C "	"	40	20,19	"	"	"	"
H27	8A "	"	36	37,18,38	"	"	"	"
H28	8B "	"	37	36,18,38	"	"	"	"
H29	8C "	"	18	36,37,38	"	"	"	"
H30	8D "	"	38	36,37,18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: Since item X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 11. Input current (high-level and low-level) test circuit for tables VIII and IX.

Source: https://assist.dla.mil/ - Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

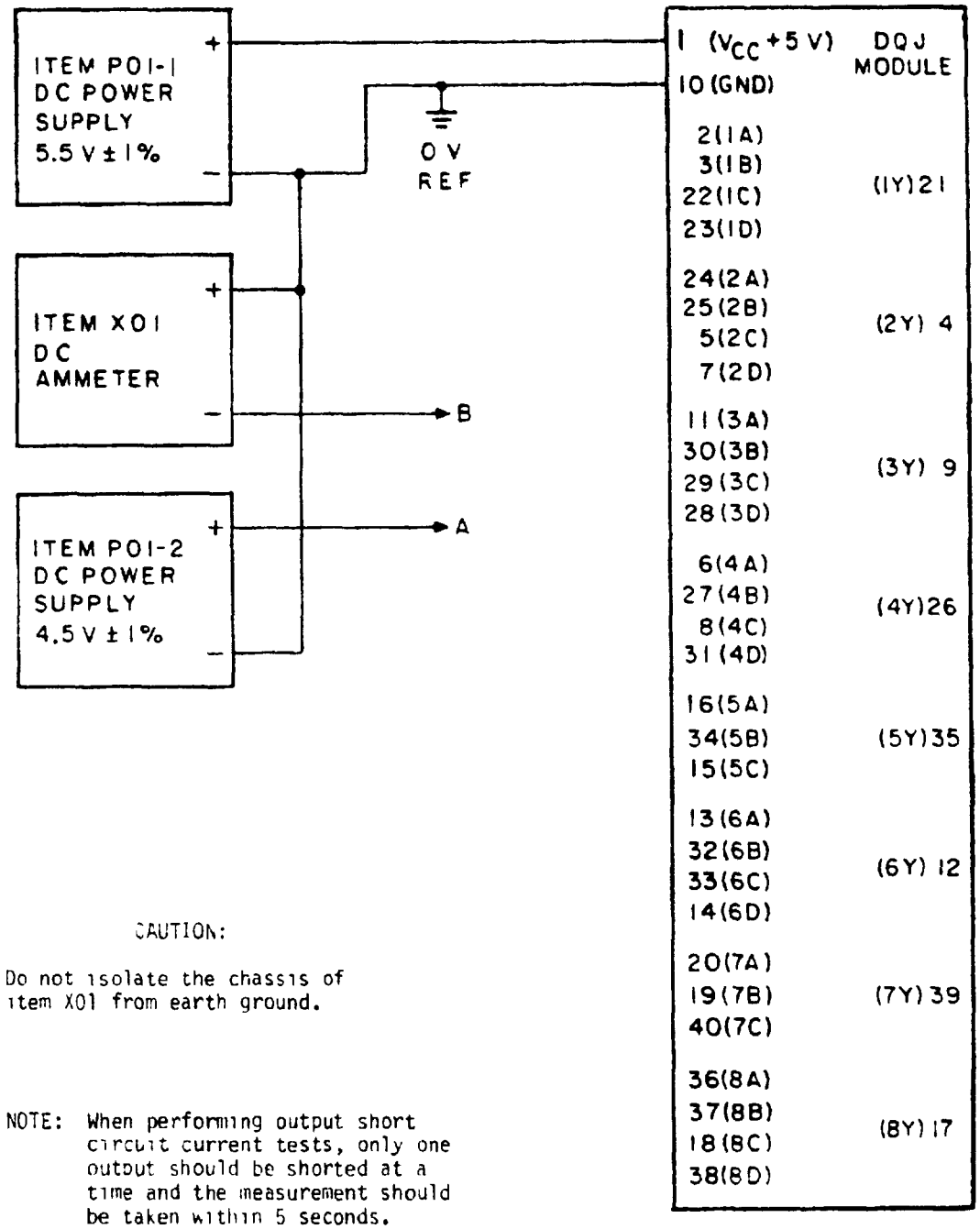
TABLE X. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1Y Output short circuit current	2,3,22,23	21	-40	-100	mA	12
J2	2Y "	24,25,5,7	4	"	"	"	"
J3	3Y "	11,30,29, 28	9	"	"	"	"
J4	4Y "	6,27,8,31	26	"	"	"	"
J5	5Y "	16,34,15	35	"	"	"	"
J6	6Y "	13,32,33, 14	12	"	"	"	"
J7	7Y "	20,19,40	39	"	"	"	"
J8	8Y "	36,37,18, 38	17	"	"	"	"

TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	Power supply current, low level	10	1	128	mA	13
K2	Power supply current, high level	B	1	80	mA	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: When performing output short circuit current tests, only one output should be shorted at a time and the measurement should be taken within 5 seconds.

FIGURE 12. Output short circuit current test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 20261020T16:55Z
Check the source to verify that this is the current version before use.

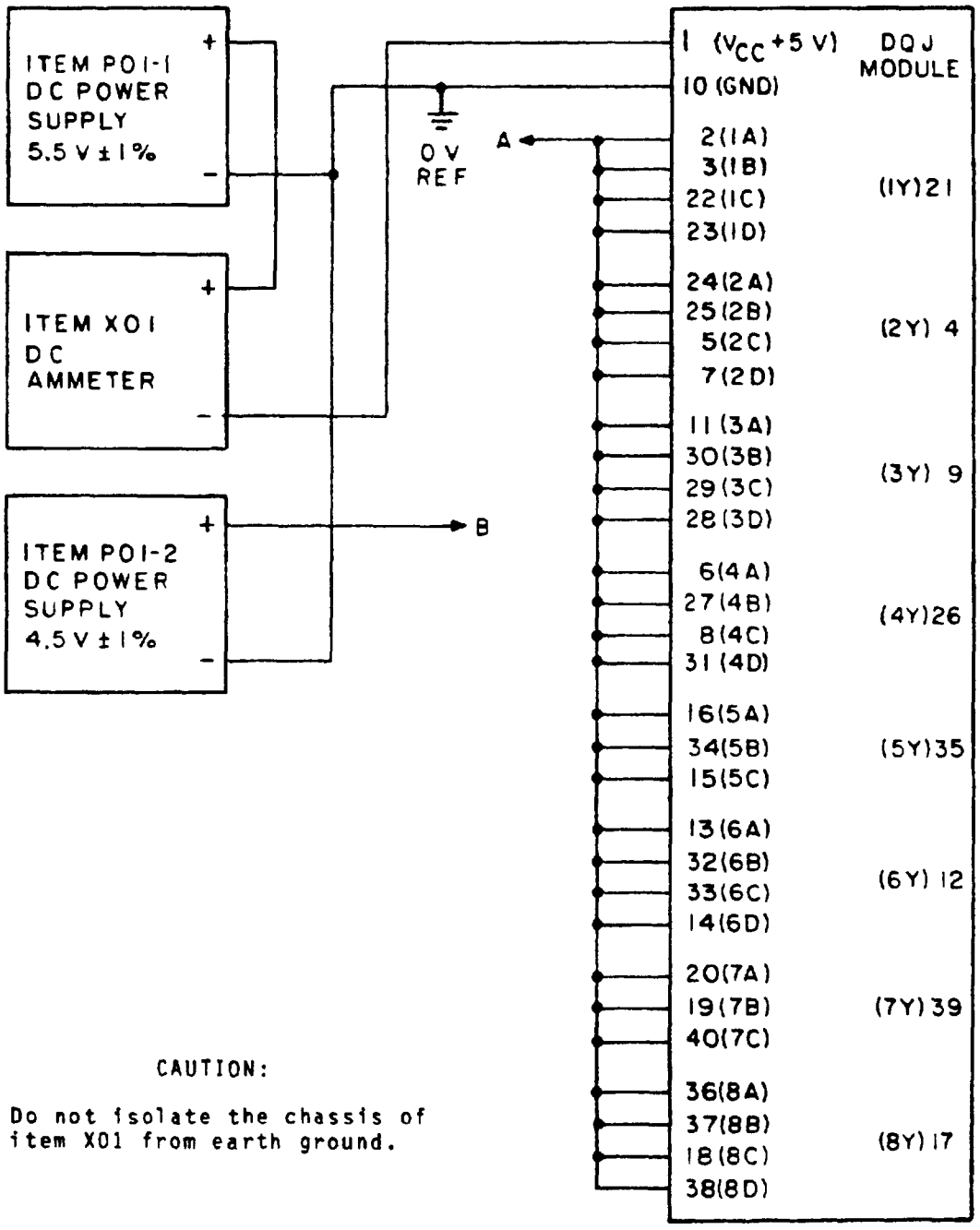


FIGURE 13. Power supply current (low-level and high-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XII. Transition time (t_{TLH}) test

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Transition time (t_{TLH}), C_L = 50 pF	2	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,18, 38	21	3	15	ns	14,15
L2	"	24	"	4	"	"	"	"
L3	"	11	"	9	"	"	"	"
L4	"	6	"	26	"	"	"	"
L5	"	16	"	35	"	"	"	"
L6	"	13	"	12	"	"	"	"
L7	"	20	"	39	"	"	"	"
L8	"	36	"	17	"	"	"	"
L9	Transition time (t_{TLH}), C_L = 150 pF	2	"	21	"	25	"	"
L10	"	24	"	4	"	"	"	"
L11	"	11	"	9	"	"	"	"
L12	"	6	"	26	"	"	"	"
L13	"	16	"	35	"	"	"	"
L14	"	13	"	12	"	"	"	"
L15	"	20	"	39	"	"	"	"
L16	"	36	"	17	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Transition time (t_{THL}), C_L = 50 pF	2	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,12, 38	21	3	15	ns	14,15
M2	"	24	"	4	"	"	"	"
M3	"	11	"	9	"	"	"	"
M4	"	6	"	26	"	"	"	"
M5	"	16	"	35	"	"	"	"
M6	"	13	"	12	"	"	"	"
M7	"	20	"	39	"	"	"	"
M8	"	36	"	17	"	"	"	"
M9	Transition time (t_{THL}), C_L = 150 pF	2	"	21	"	25	"	"
M10	"	24	"	4	"	"	"	"
M11	"	11	"	9	"	"	"	"
M12	"	6	"	26	"	"	"	"
M13	"	16	"	35	"	"	"	"
M14	"	13	"	12	"	"	"	"
M15	"	20	"	39	"	"	"	"
M16	"	36	"	17	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Propagation delay time (t_{PLH}), C_L = 50 pF	2	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,18, 38	21	3	15	ns	14,15
N2	"	24	"	4	"	"	"	"
N3	"	11	"	9	"	"	"	"
N4	"	6	"	26	"	"	"	"
N5	"	16	"	35	"	"	"	"
N6	"	13	"	12	"	"	"	"
N7	"	20	"	39	"	"	"	"
N8	"	36	"	17	"	"	"	"
N9	"	3	2,24,11,6,16,13, 20,36,22,5,29, 8,15,33,40,18, 23,7,28,31,14, 38	21	"	"	"	"
N10	"	25	"	4	"	"	"	"
N11	"	30	"	9	"	"	"	"
N12	"	27	"	26	"	"	"	"
N13	"	34	"	35	"	"	"	"
N14	"	32	"	12	"	"	"	"
N15	"	19	"	39	"	"	"	"
N16	"	37	"	17	"	"	"	"
N17	"	22	"	21	"	"	"	"

Source: https://assist.dla.mil - Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N18	Propagation delay time (t_{PLH}), C_L = 50 pF	5	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 23,7,28,31,14, 38	4	3	15	ns	14,15
N19	"	29	"	9	"	"	"	"
N20	"	8	"	26	"	"	"	"
N21	"	15	"	35	"	"	"	"
N22	"	33	"	12	"	"	"	"
N23	"	40	"	39	"	"	"	"
N24	"	18	"	17	"	"	"	"
N25	"	23	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 22,5,29,8,15,33, 40,18	21	"	"	"	"
N26	"	7	"	4	"	"	"	"
N27	"	28	"	9	"	"	"	"
N28	"	31	"	26	"	"	"	"
N29	"	14	"	12	"	"	"	"
N30	"	38	"	17	"	"	"	"
N31	Propagation delay time (t_{PLH}), C_L = 150 pF	2	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,18, 38	21	"	25	"	"
N32	"	24	"	4	"	"	"	"
N33	"	11	"	9	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N34	Propagation delay time (t_{PLH}), C_L = 150 pF	6	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,18, 38	26	3	25	ns	14,15
N35	"	16	"	35	"	"	"	"
N36	"	13	"	12	"	"	"	"
N37	"	20	"	39	"	"	"	"
N38	"	36	"	17	"	"	"	"
N39	"	3	2,24,11,6,16,13, 20,36,22,5,29, 8,15,33,40,18, 23,7,28,31,14, 38	21	"	"	"	"
N40	"	25	"	4	"	"	"	"
N41	"	30	"	9	"	"	"	"
N42	"	27	"	26	"	"	"	"
N43	"	34	"	35	"	"	"	"
N44	"	32	"	12	"	"	"	"
N45	"	19	"	39	"	"	"	"
N46	"	37	"	17	"	"	"	"
N47	"	22	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 23,7,28,31,14, 38	21	"	"	"	"
N48	"	5	"	4	"	"	"	"
N49	"	29	"	9	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XIV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N50	Propagation delay time (t_{PLH}), C_L = 150 pF	8	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 23,7,28,31,14, 38	26	3	25	ns	14,15
N51	"	15	"	35	"	"	"	"
N52	"	33	"	12	"	"	"	"
N53	"	40	"	39	"	"	"	"
N54	"	18	"	19	"	"	"	"
N55	"	23	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 22,5,29,8,15,33, 40,18	21	"	"	"	"
N56	"	7	"	4	"	"	"	"
N57	"	28	"	9	"	"	"	"
N58	"	31	"	26	"	"	"	"
N59	"	14	"	12	"	"	"	"
N60	"	38	"	17	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Propagation delay time (t_{PHL}), C_L = 50 pF	2	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,18, 38	21	3	15	ns	14,15
P2	"	24	"	4	"	"	"	"
P3	"	11	"	9	"	"	"	"
P4	"	[illegible]	"	26	"	"	"	"
P5	"	16	"	35	"	"	"	"
P6	"	12	"	12	"	"	"	"
P7	"	20	"	39	"	"	"	"
P8	"	36	"	17	"	"	"	"
P9	"	3	2,24,11,6,16,13, 20,36,22,5,29, 8,15,33,40,18, 23,7,28,31,14, 38	21	"	"	"	"
P10	"	25	"	4	"	"	"	"
P11	"	30	"	9	"	"	"	"
P12	"	27	"	26	"	"	"	"
P13	"	34	"	35	"	"	"	"
P14	"	32	"	12	"	"	"	"
P15	"	19	"	39	"	"	"	"
P16	"	37	"	17	"	"	"	"
P17	"	22	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 23,7,28,31,14, 38	21	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P18	Propagation delay time (t_{PHL}), C_L = 50 pF	5	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 23,7,28,31,14, 38	4	3	15	ns	14,15
P19	"	29	"	9	"	"	"	"
P20	"	8	"	26	"	"	"	"
P21	"	15	"	35	"	"	"	"
P22	"	33	"	12	"	"	"	"
P23	"	40	"	39	"	"	"	"
P24	"	18	"	17	"	"	"	"
P25	"	23	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 22,5,29,8,15,33, 40,18	21	"	"	"	"
P26	"	7	"	4	"	"	"	"
P27	"	28	"	9	"	"	"	"
P28	"	31	"	26	"	"	"	"
P29	"	14	"	12	"	"	"	"
P30	"	38	"	17	"	"	"	"
P31	Propagation delay time (t_{PHL}), C_L = 150 pF	2	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,18, 38	21	"	25	"	"
P32	"	24	"	4	"	"	"	"
P33	"	11	"	9	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

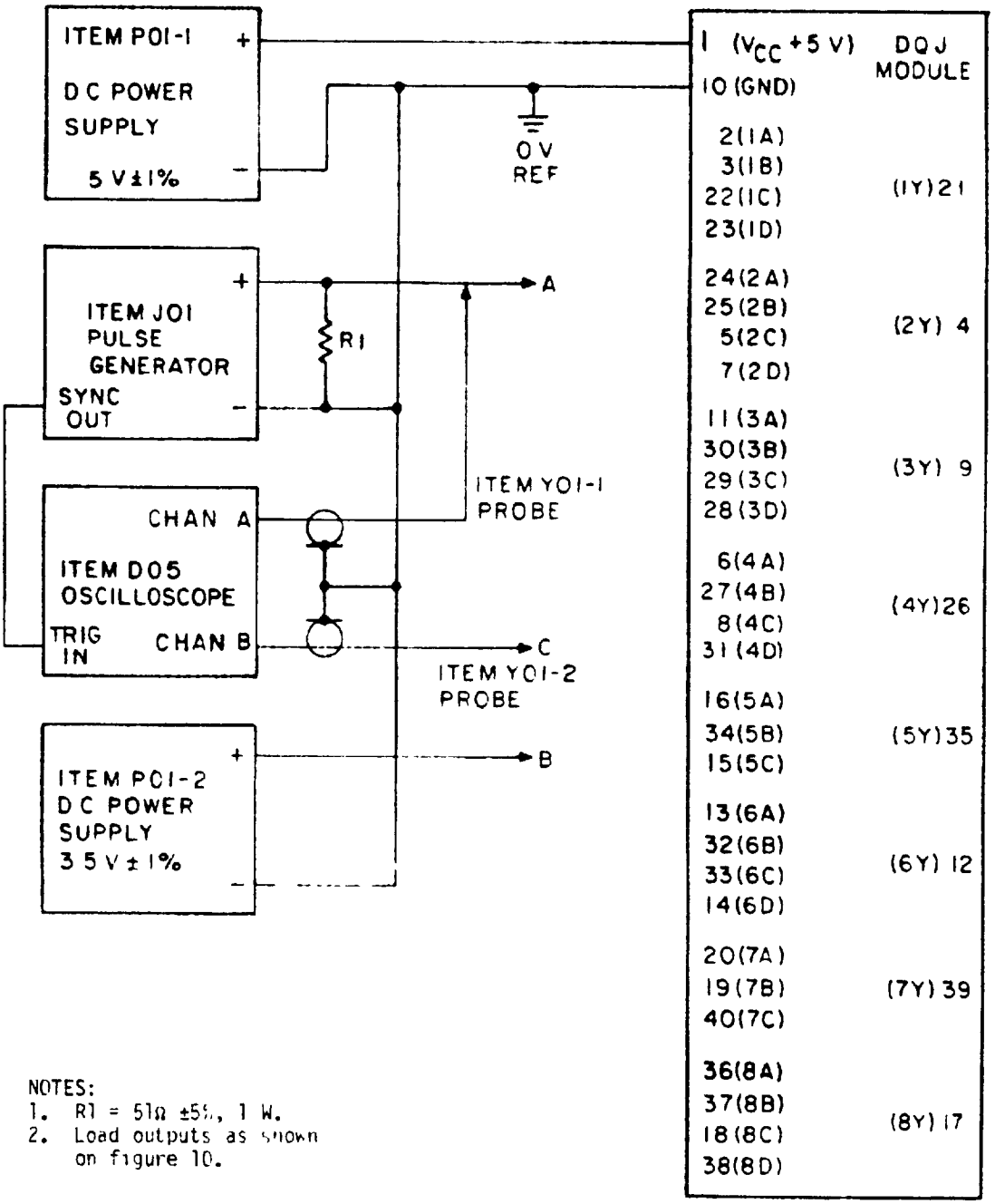
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P34	Propagation delay time (t_{PHL}), C_L = 150 pF	6	3,22,23,25,5,7, 30,29,28,27,8, 31,34,15,32,33, 14,19,40,37,18, 38	26	3	25	ns	14,15
P35	"	16	"	35	"	"	"	"
P36	"	13	"	12	"	"	"	"
P37	"	[illegible]	"	39	"	"	"	"
P38	"	[illegible]	"	17	"	"	"	"
P39	"	3	2,24,11,6,16,13, 20,36,22,5,29, 8,15,33,40,18, 23,7,28,31,14, 38	21	"	"	"	"
P40	"	25	"	4	"	"	"	"
P41	"	30	"	9	"	"	"	"
P42	"	27	"	26	"	"	"	"
P43	"	34	"	35	"	"	"	"
P44	"	32	"	12	"	"	"	"
P45	"	19	"	39	"	"	"	"
P46	"	37	"	17	"	"	"	"
P47	"	22	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 23,7,28,31,14, 38	21	"	"	"	"
P48	"	5	"	4	"	"	"	"
P49	"	29	"	9	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P50	Propagation delay time (t_{PHL}), C_L = 150 pF	8	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 23,7,28,31,14, 38	26	3	25	ns	14,15
P51	"	15	"	35	"	"	"	"
P52	"	33	"	12	"	"	"	"
P53	"	40	"	39	"	"	"	"
P54	"	18	"	17	"	"	"	"
P55	"	23	2,24,11,6,16,13, 20,36,3,25,30, 27,34,32,19,37, 22,5,29,8,15,33, 40,18	21	"	"	"	"
P56	"	7	"	4	"	"	"	"
P57	"	28	"	9	"	"	"	"
P58	"	31	"	26	"	"	"	"
P59	"	14	"	12	"	"	"	"
P60	"	38	"	17	"	"	"	"

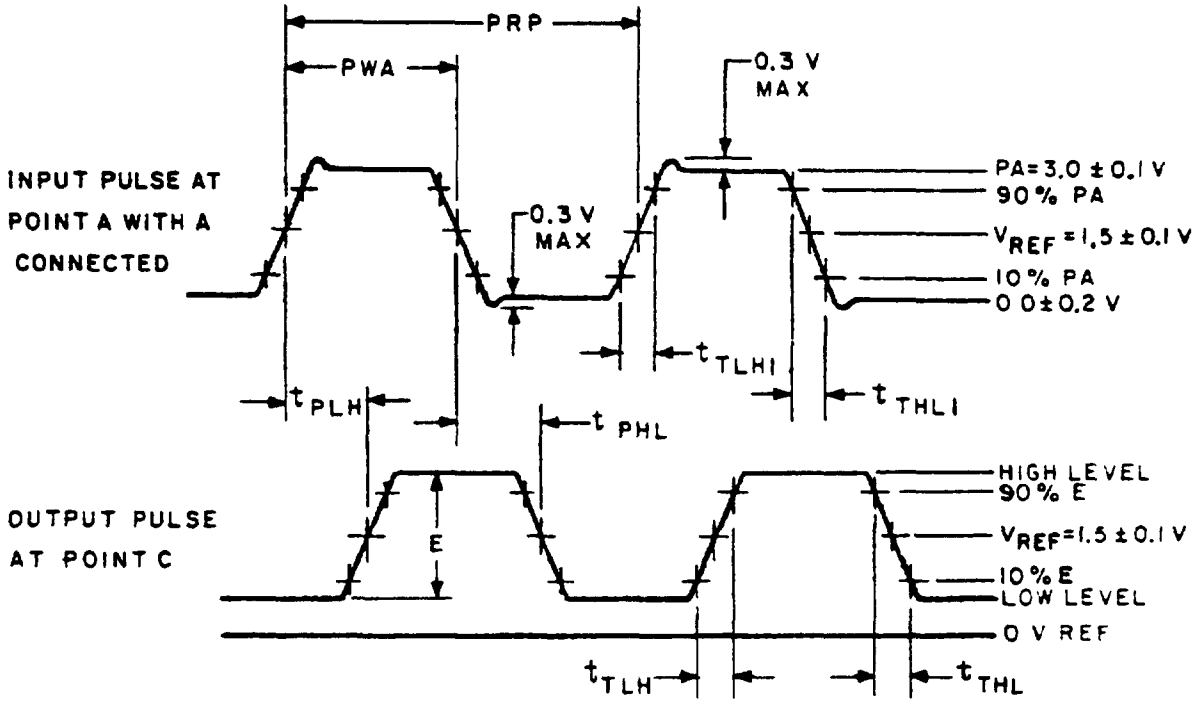
Source: https://assist.dla.mil -- Downloaded: 20261O-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load outputs as shown on figure 10.

FIGURE 14 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII, XIV, and XV

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins, items YO1-1 and YO1-2 to output pins; and between load circuits and output pins shall be less than 2 inches (50.8 mm).
4. All voltages are referenced to ground.

FIGURE 15 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII, XIV, and XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low-level) test

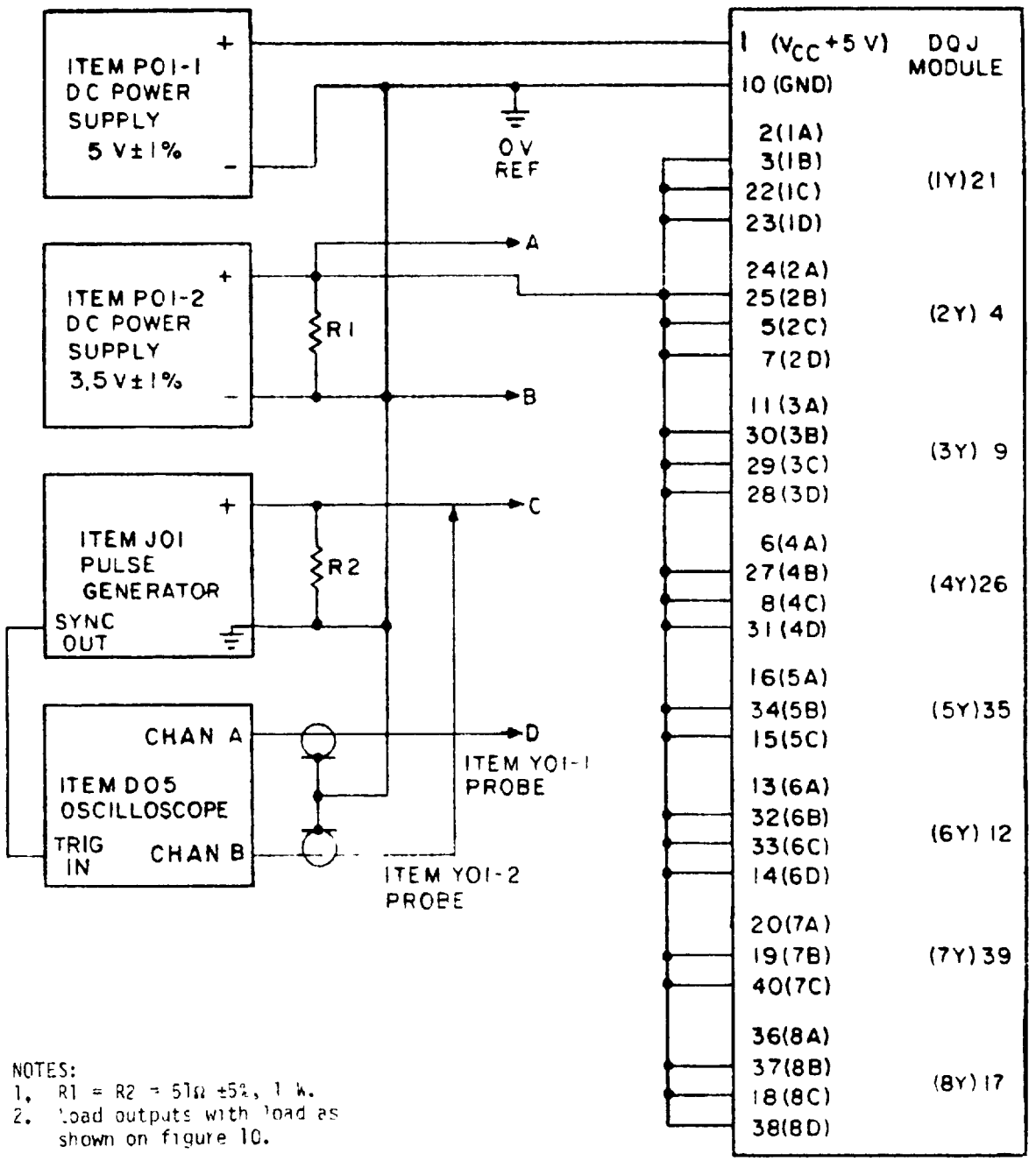
Test condition		Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Y1	Output crosstalk, low level	Open	2	24,11,6,16, 13,20,36	21	0	400	mV	16,17
Q2	Y2	"	"	24	2,11,6,16, 13,20,36	4	"	"	"	"
Q3	Y3	"	"	11	2,24,6,16, 13,20,36	9	"	"	"	"
Q4	Y4	"	"	6	2,24,11,16, 13,20,36	26	"	"	"	"
Q5	Y5	"	"	16	2,24,11,6, 13,20,36	35	"	"	"	"
Q6	Y6	"	"	13	2,24,11,6, 16,20,36	12	"	"	"	"
Q7	Y7	"	"	20	2,24,11,6, 16,13,36	39	"	"	"	"
Q8	Y8	"	"	36	2,24,11,6, 16,13,20	17	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2022-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Y1	Output crosstalk, high level	2	Open	24,11,6,16, 13,20,36	21	0	400	mV	16,17
R2	Y2	"	24	"	2,11,6,16, 13,20,36	4	"	"	"	"
R3	Y3	"	11	"	2,24,6,16, 13,20,36	9	"	"	"	"
R4	Y4	"	6	"	2,24,11,16, 13,20,36	26	"	"	"	"
R5	Y5	"	16	"	2,24,11,6, 13,20,36	35	"	"	"	"
R6	Y6	"	13	"	2,24,11,6, 16,20,36	12	"	"	"	"
R7	Y7	"	20	"	2,24,11,6, 16,13,36	39	"	"	"	"
R8	Y8	"	36	"	2,24,11,6, 16,13,20	17	"	"	"	"

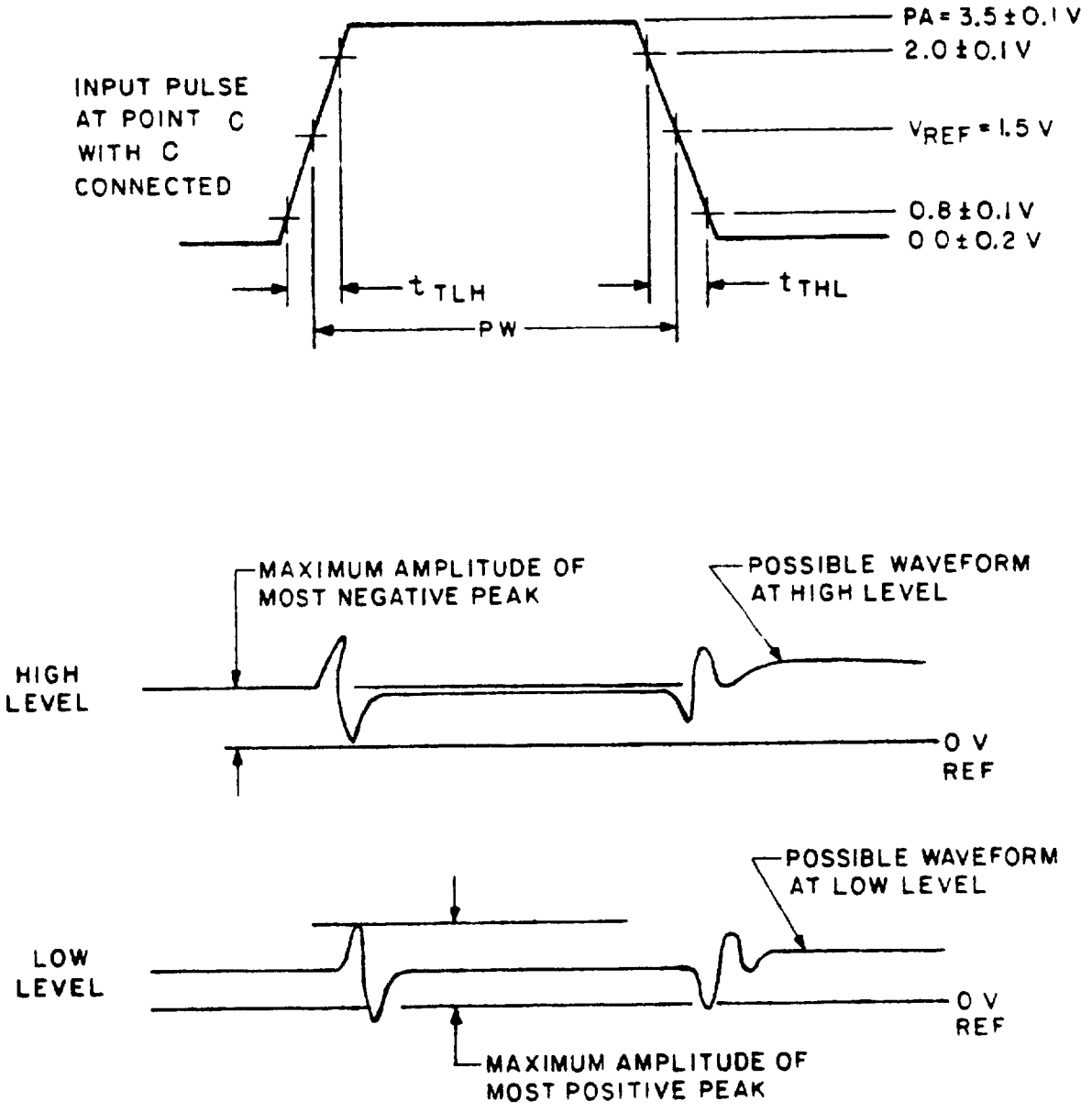
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load outputs with load as shown on figure 10.

FIGURE 16 Output crosstalk (low-level and high-level) test circuit for tables XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTE Input pulse parameters
PA = 3.5 ±0.1 V.
PW = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1 MHz ±10%

FIGURE 17 Output crosstalk (low-level and high-level) test circuit waveforms for tables XVI and XVII

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.

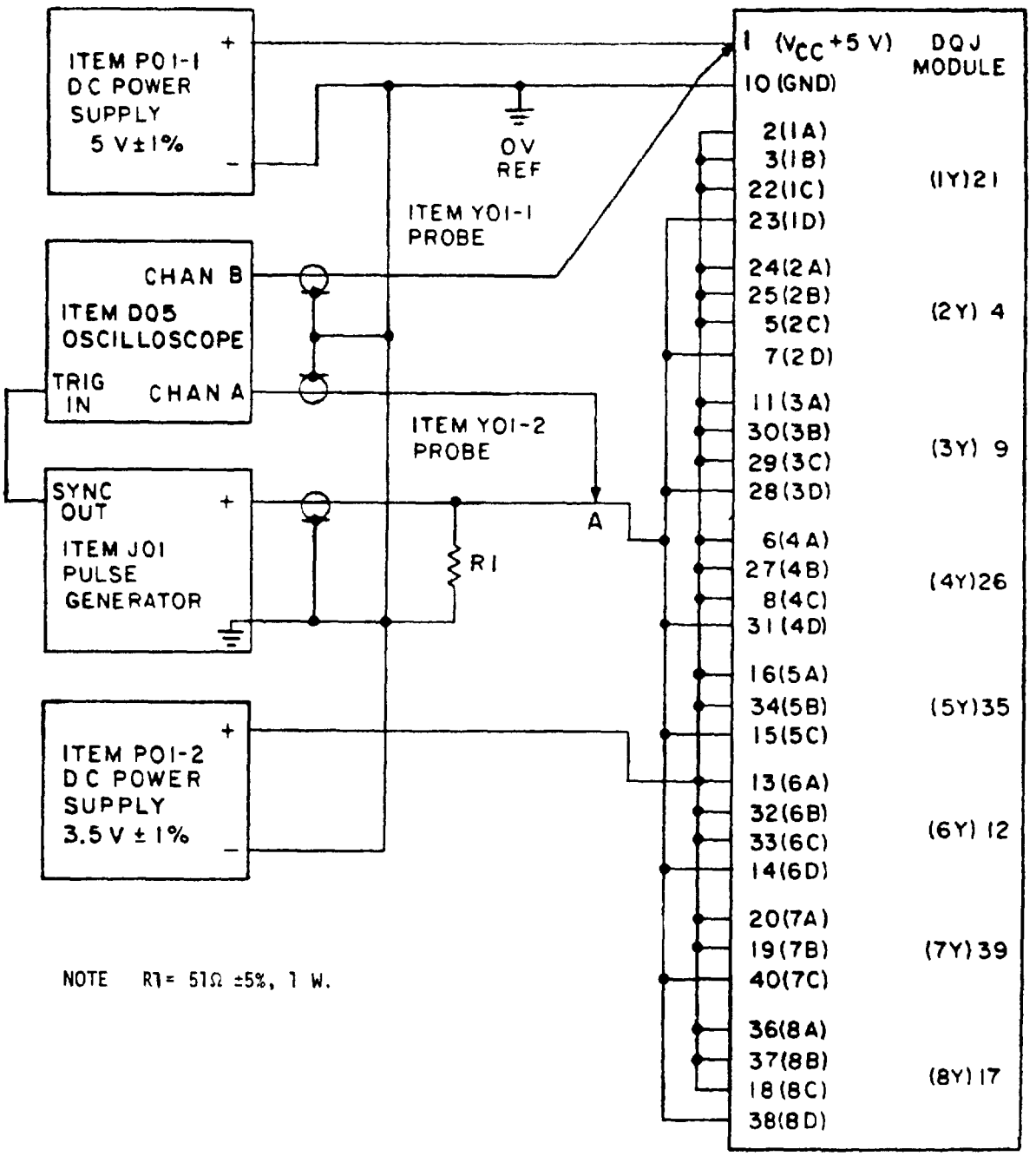
TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	400	mV	18,19

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T	Power supply filter capacitance	1	10	1.6	3.0	µF	20

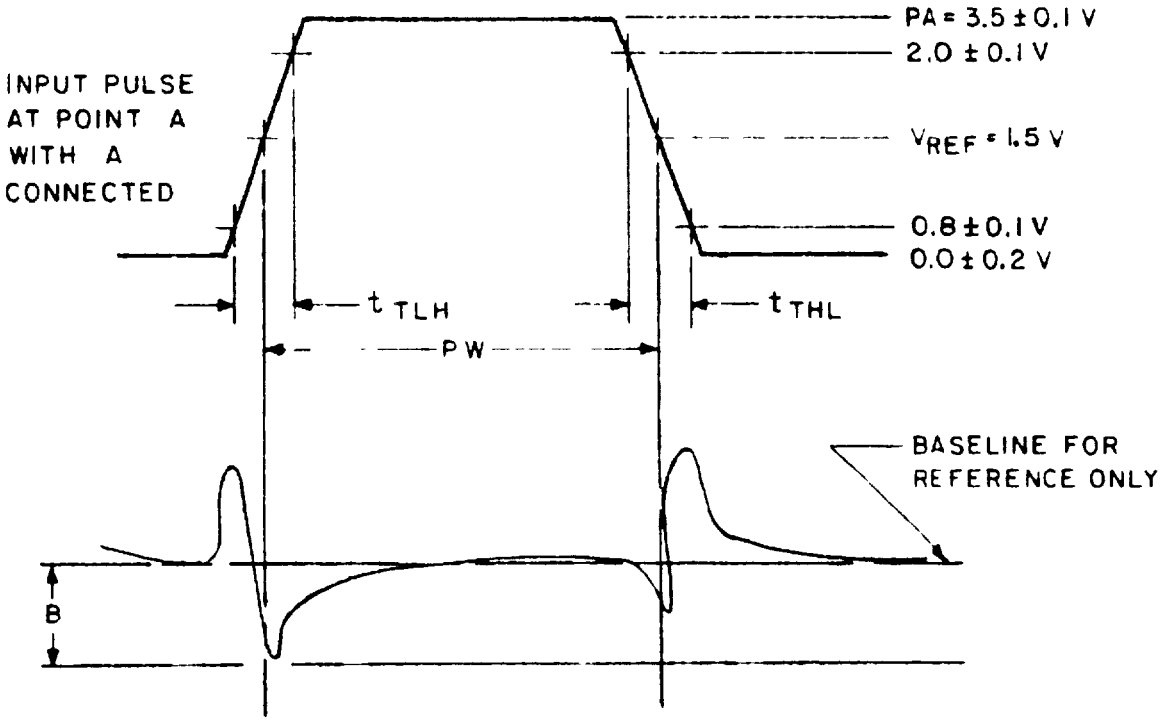
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTE R1 = 51Ω ±5%, 1 W.

* FIGURE 18. Power supply transient amplitude test circuit for table XVIII.

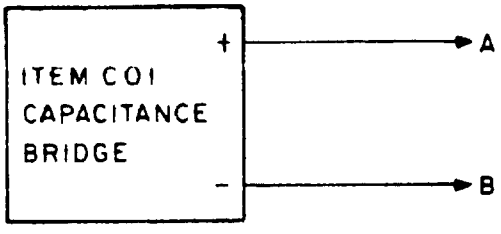
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters
   PA = 3.5 ±0.1 V.
   PW = 3 µs ±10%.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns verified at each input pin.
   PRR = 100 kHz ±10%.
2. B = Measurement from the baseline to the most negative excursion of the waveform.

FIGURE 19 Power supply transient amplitude test circuit waveforms for table XVIII

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.



1 (V_{CC} +5 V) 10 (GND)	DQJ MODULE
2(1A) 3(1B) 22(1C) 23(1D)	(1Y)21
24(2A) 25(2B) 5(2C) 7(2D)	(2Y) 4
11(3A) 30(3B) 29(3C) 28(3D)	(3Y) 9
6(4A) 27(4B) 8(4C) 31(4D)	(4Y)26
16(5A) 34(5B) 15(5C)	(5Y)35
13(6A) 32(6B) 33(6C) 14(6D)	(6Y) 12
20(7A) 19(7B) 40(7C)	(7Y) 39
36(8A) 37(8B) 18(8C) 38(8D)	(8Y) 17

NOTE Item C01 setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

* FIGURE 20. Power supply filter capacitance test circuit for table XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XX. Input capacitance test.

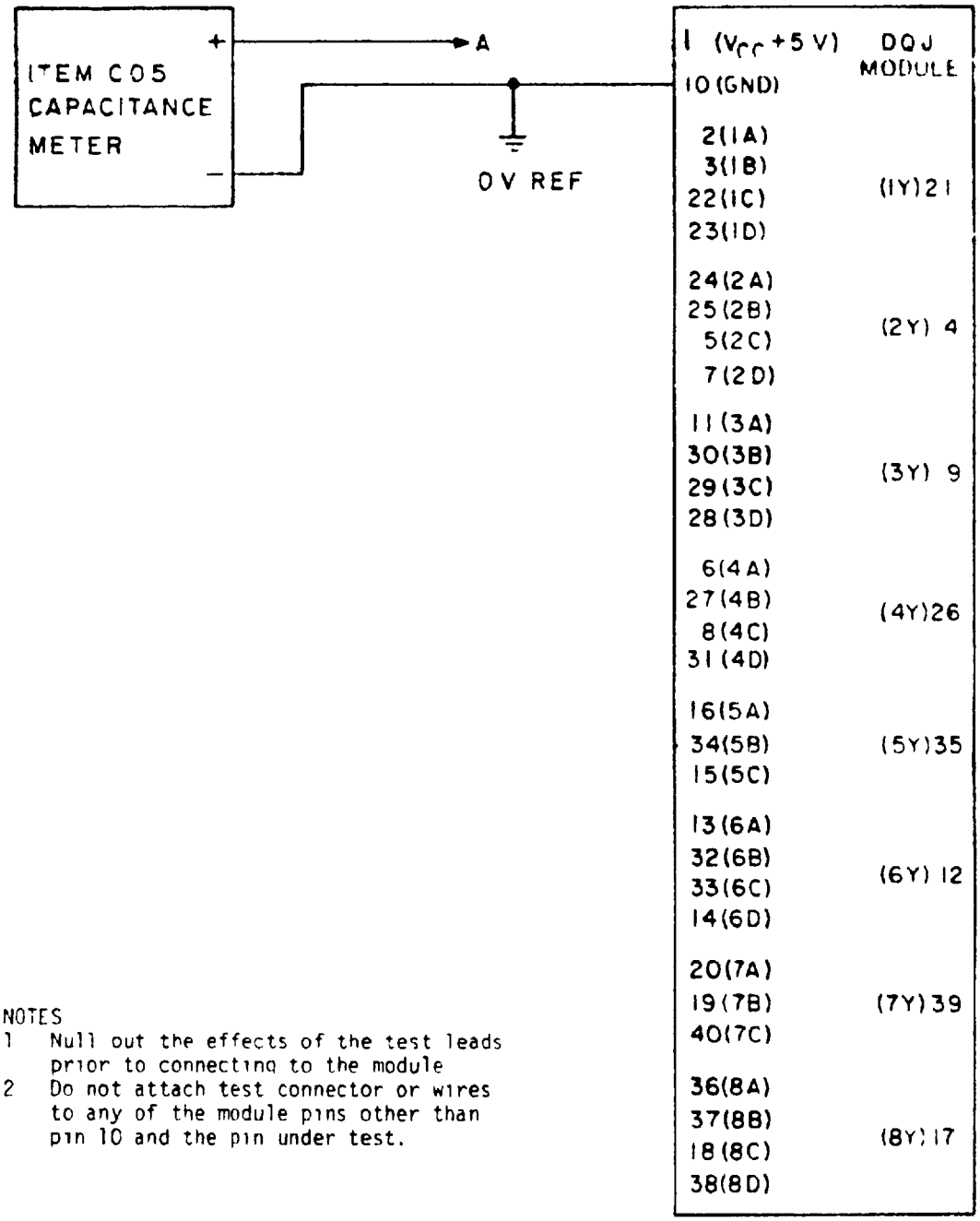
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
U1	1A Input capacitance	2	2	15	pF	21
U2	1B "	3	"	"	"	"
U3	1C "	22	"	"	"	"
U4	1D "	23	"	"	"	"
U5	2A "	24	"	"	"	"
U6	2B "	25	"	"	"	"
U7	2C "	5	"	"	"	"
U8	2D "	7	"	"	"	"
U9	3A "	11	"	"	"	"
U10	3B "	30	"	"	"	"
U11	3C "	29	"	"	"	"
U12	3D "	28	"	"	"	"
U13	4A "	6	"	"	"	"
U14	4B "	27	"	"	"	"
U15	4C "	8	"	"	"	"
U16	4D "	31	"	"	"	"
U17	5A "	16	"	"	"	"
U18	5B "	34	"	"	"	"
U19	5C "	15	"	"	"	"
U20	6A "	13	"	"	"	"
U21	6B "	32	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T16:55Z
Check the source to verify that this is the current version before use.

TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U22	6C Input capacitance	33	2	15	pF	21
U23	6D "	14	"	"	"	"
U24	7A "	20	"	"	"	"
U25	7B "	19	"	"	"	"
U26	7C "	40	"	"	"	"
U27	8A "	36	"	"	"	"
U28	8B "	37	"	"	"	"
U29	8C "	18	"	"	"	"
U30	8D "	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:55Z
Check the source to verify that this is the current version before use.



NOTES
1 Null out the effects of the test leads prior to connecting to the module
2 Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test.

FIGURE 21 Input capacitance test circuit for table XX

GOVERNMENT PRINTING OFFICE 1989 241-048/1485
Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/244A(EC)	MODULES, STANDARD ELECTRONIC GATE, AND, DIGITAL, KEY CODE DQJ

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-04-20 11:10:33
Check the source to verify that this is the current version before use.